

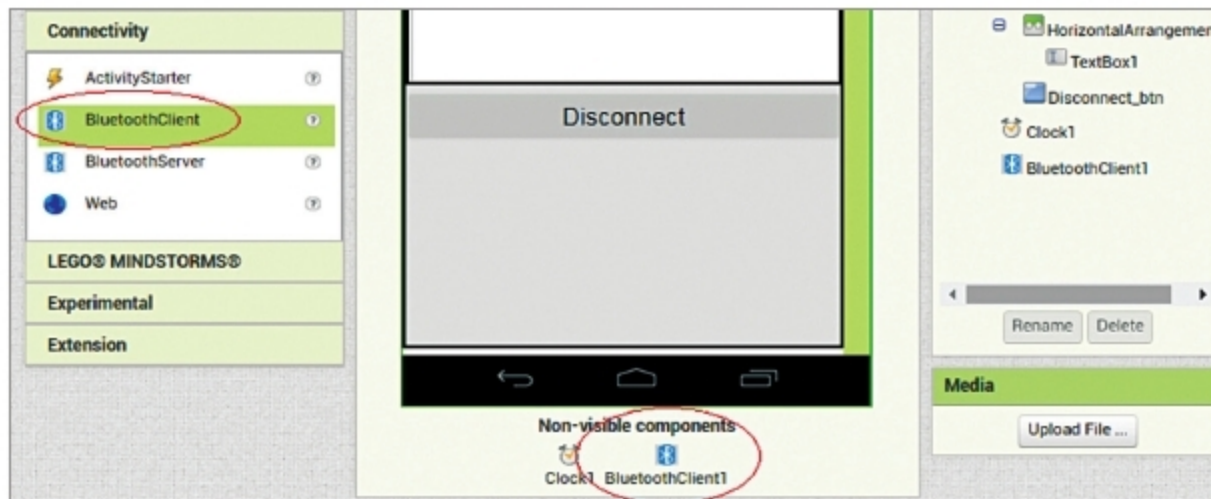


Figure 13: Adding Bluetooth Client

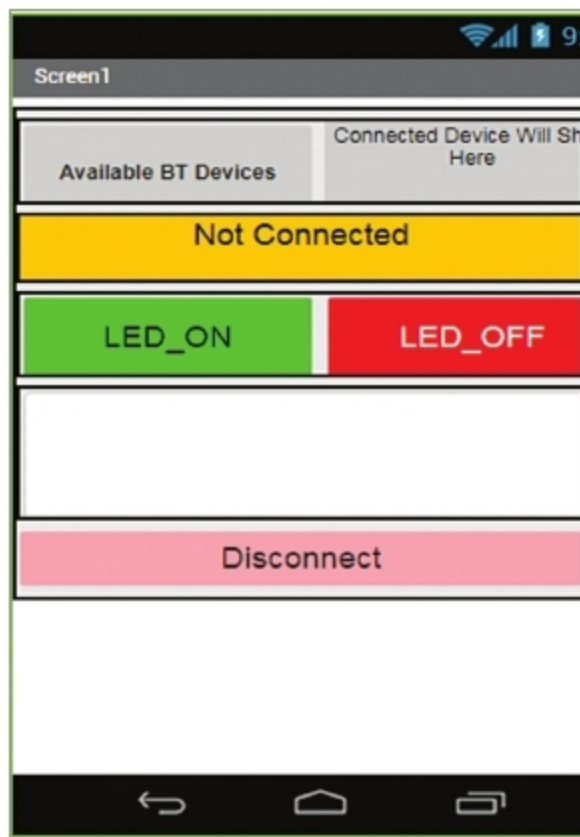


Figure 14: Final user interface on mobile

following changes:

- Height: fill parent
 - Width: fill parent
 - Hint: serial data will show here
- Adding disconnect button:** Select *Palette*→*User Interface*→*Button* and drag it over *Vertical Arrangement1*. Go to *Properties* of *Button* and make the following changes:
- Background colour: pink
 - Font size: 19
 - Height: automatic
 - Width: fill parent
 - Text: disconnect
 - Text alignment: centre
- Non-visible component:** Now, add some non-visible components. Select *Palette* → *Sensor* → *Clock* and drag it to *Screen1*.

Adding Bluetooth Client: Bluetooth Client component is used to establish Bluetooth connection. Select

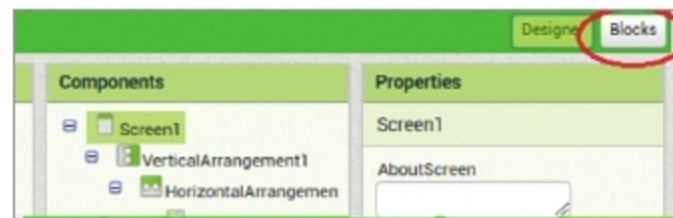


Figure 15: Blocks

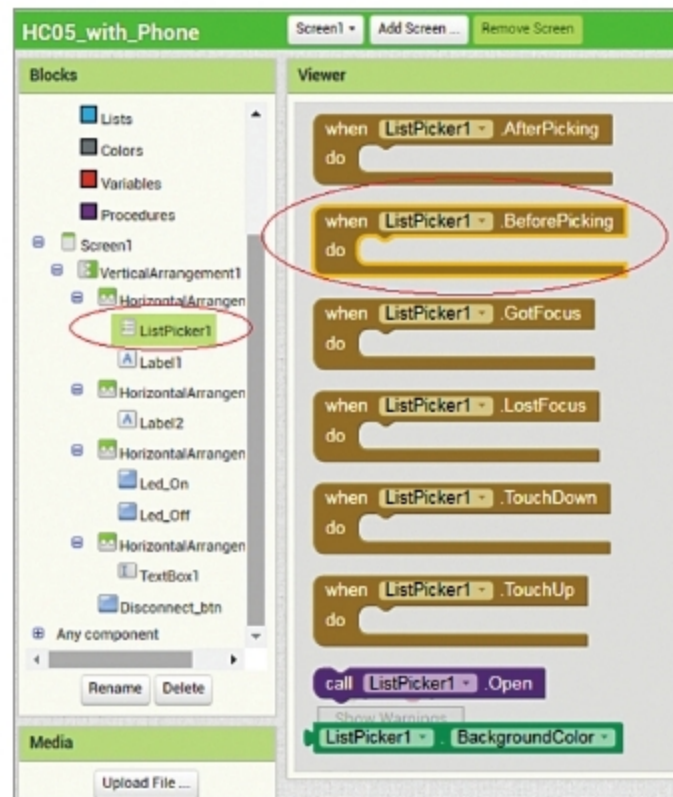


Figure 16: Viewer window

Palette→*Connectivity*→*Bluetooth Client* and drag it to *Screen1*.

Finally, the app will look as shown in Figure 14.

App Inventor Blocks section

Now, let us define the working of each component as per our requirement. *Blocks* section (Figure 15) is where you provide the functionality to your selected components. There are built-in blocks that can handle things like Maths, logic and text, which you can use with each component that you have added. In addition to these components, the project has its own set of blocks specific to its own events, methods and properties.

Click on *Blocks* at the top-right corner in *Screen1*. You will get a new window. Click on *ListPicker1*, as shown in Figure 16. Select *ListPicker1 BeforePicking* event, and drag and drop it over *Viewer* window.

From *ListPicker1*, select *Set ListPicker1.Elements To*, and drag and drop it over *Viewer* window. Then, go to *BluetoothClient1* under *Screen1* and select *BluetoothClient1.AddressAndNames*, and drag and drop it over *Viewer* window. Arrange all events in a similar fashion, as shown in Figure 17.

BeforePicking(): This is the event to raise when *Button* is clicked. The event occurs before the list of items is displayed. It is used to prepare the list before it is shown. It sets the elements of *ListPicker* to available Bluetooth devices by their name and address.

AfterPicking(): This event is to be raised after the picker activity returns its result and properties have been filled in. Select *ListPicker1.AfterPicking* and go to built-in block *Logic*. Select another *if* then event, as shown in Figure 17.

If the 'if' condition is satisfied or true, function *BluetoothClient1* will be connected to a device from *ListPicker* section. It will set *label1* text as *Connected BT Device* and *label2* text as *Connected*, and change the text colour to green.

Now, add blocks for *LED_ON* and *LED_OFF* buttons, as shown in Figure 18. Here, send 1 to turn on the LED and 0 to turn it off. You can also send characters or strings according to the code.

Next, add an event for *LED_ON* button. When this button is clicked, call *BluetoothClient1.SendText* function to send text as 1. Do the same for *LED_OFF*; send 0 on clicking.

Adding clock: Select an event from *Clock1.Timer*, as shown in Figure 19. Here, clock is used to receive data from the client.

From *Blocks* on the left side, select 'and' logic and drag it over *Viewer* window. This returns true if both conditions are satisfied. Then, select *BluetoothClient1.IsConnected* and attach it to 'and' logic to one side. Now, go to *Math*, select '=' logic and select greater than (>) symbol.